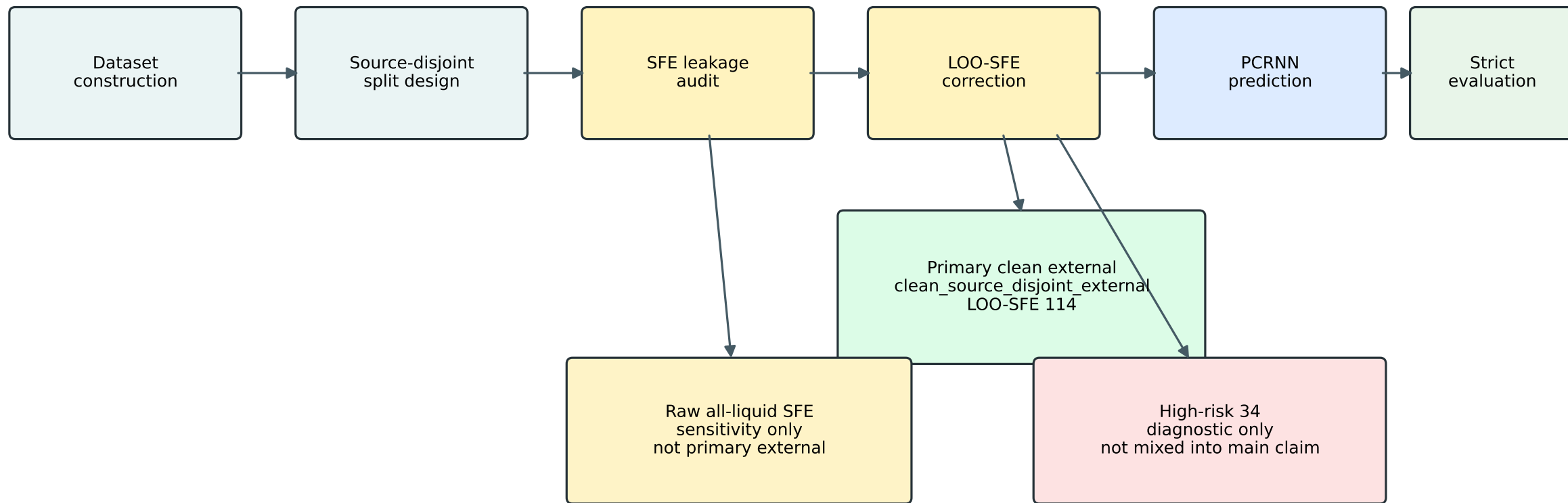


Leakage-aware physics-constrained contact-angle prediction workflow



Key rule: the clean external claim uses LOO-SFE 114; all-liquid SFE and high-risk 34 are separated as sensitivity/diagnostic evidence.